

Scikit - Learn :

sklearn is a python library for ML that provides, ML algorithms, preprocessing tools, evaluation metrics, model selection utilities.

- It is built on top of numpy, scipy and matplotlib

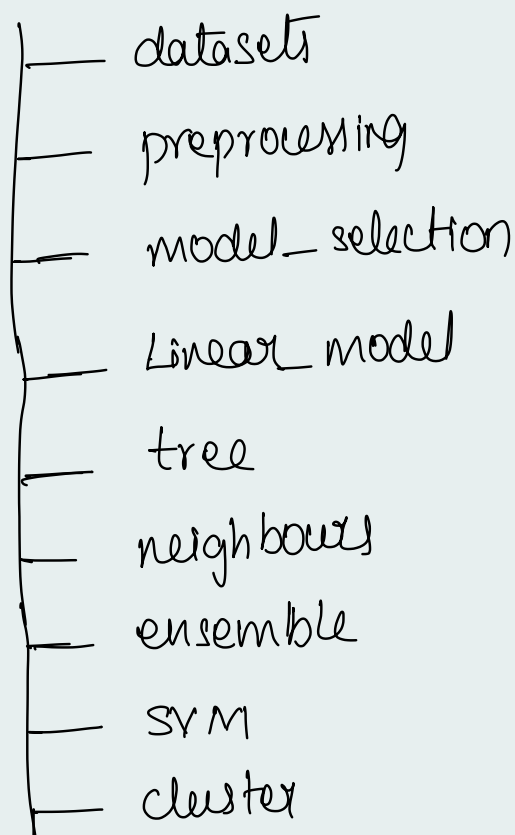
Simple pipeline:

Dataset → Preprocessing → Model → Prediction → Evaluation

sklearn provides tools for each step.

High-level structure of sklearn:

sklearn



- metrics
- decomposition
- pipeline.

— Each module handles a specific ML task.

Datasets → sklearn.datasets

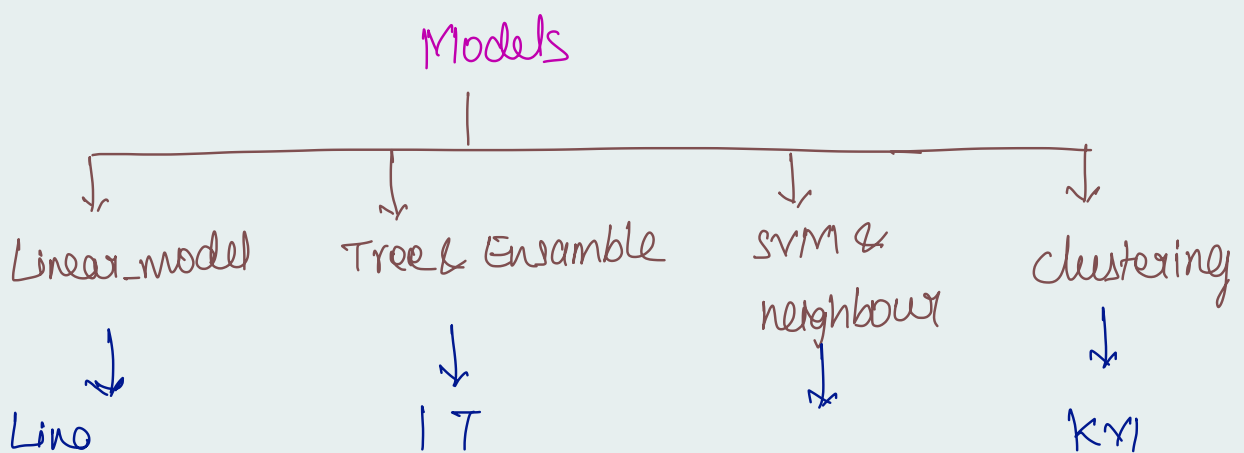
- load_iris()
- load_boston()
- load_classification()

Preprocessing → sklearn.preprocessing

- StandardScaler
- MinMaxScaler
- OneHotEncoder
- LabelEncoder

Model_selection → sklearn.model_selection

- train_test_split
- GridSearchCV
- cross_val_score

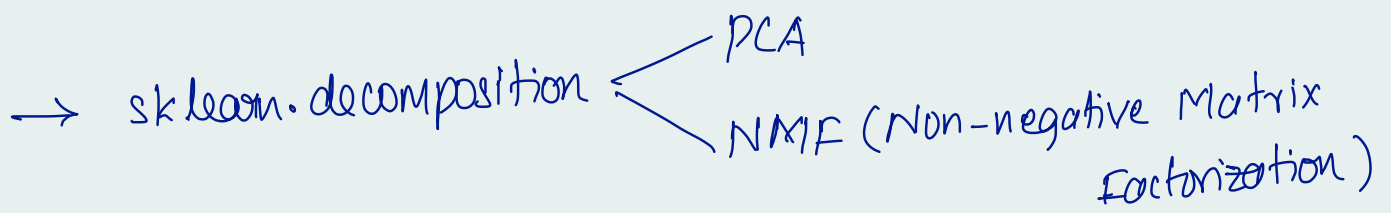


Logistic Reg

RF

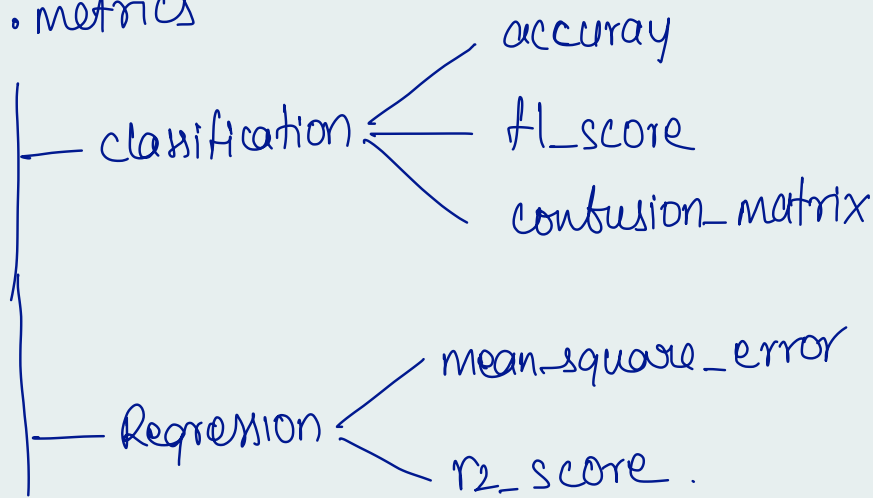
kneighbour

DBSCAN



Evaluation Metrics:

sklearn.metrics



Scikit-learn is a toolbox where each module handles one stage of the ML pipeline.

